CDS524 Group Project: Machine Learning Application

1. Team Information

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1.1 Contribution Statement

Liu Ruiyang: Controlled variable experiment design, report writing.

Xing Jiayi: Project design, XGBoost implementation.

Xia Meng: LSTM implementation, hyperparameter tuning, model evaluation.

Zhou Bojian: Data preprocessing, EDA, feature engineering (Chinese cosmological features).

Xue Tingkun: Statistical analysis, ablation study, presentation preparation.

1.2 GitHub Link

https://github.com/zhoubojian-stevenchow/CDS524-Group-Project

2. Problem Statement

This project proposes next-day temperature forecasting for European cities using machine learning. Short-term temperature forecasting is essential for agriculture, energy management, transportation planning, and public health early warning systems. It is framed as a time-series regression problem: given today's weather observations for a city, forecast tomorrow's mean, min, and max temperatures.

We compare two fundamentally different modeling paradigms: XGBoost (gradient boosting on tabular features) and bidirectional LSTM with Attention (deep learning on sequential data). Both models are trained independently for each city, with three target predictions per city (temp_mean, temp_min, and temp_max) for five European cities, resulting in 15 city-target evaluation pairs per model.

In addition to the primary model comparison, we perform a 2×2 factorial controlled variable experiment to isolate the effects of two feature engineering strategies: (1) lag and rolling statistical features, and (2) Chinese cosmological features, including 24 Solar Terms, Wuxing/Five Elements, and Heavenly Stems. Such an experimental design enables us to decide whether these features contribute predictive power or only add redundancy and noise.

3. Dataset

3.1 Source and Description

Dataset: European Cities Weather Prediction Dataset

URL:
https://www.kaggle.com/datasets/orvile/european-cities-weather-prediction-dataset

Original source: European Climate Assessment & Dataset (ECA&D) [6].

The data set includes 18 European cities over ten years, from 1 January 2000 to 1 January 2010, and consists of 3,654 daily weather observations. Each city was endowed with a maximum of 11 different weather features, including cloud_cover, wind_speed, wind_gust, humidity, pressure, global_radiation, precipitation, sunshine, temp_mean, temp_min, and temp_max. There are no null records in the database, and any invalid values for features amounting to less than 5% have been replaced by the respective columns in the original data set.

3.2 City Selection

The authors of the original data set introduce the following five cities: DE_BILT, DUSSELDORF, MAASTRICHT, MUENCHEN, and OSLO. All 11 weather features are available. To conduct proper experimentation and ensure comparability for the different tests, we have decided to restrict ourselves to these five cities. We have excluded Budapest, as the feature temp_min is absent, and Malmo, as it has only five out of the 11 features.

Property	Value	Property	Value	Property	Value
Total rows	3,654	Total columns	165	Date range	2000–2010

Cities (total)	18	Cities (used)	5	Null values	0

Table 1: Dataset summary statistics.

3.3 Train/Validation/Test Split

We adopt a strict chronological split to avoid temporal data leakage: The training set covers the period 2000–2007 (~2,922 days), the validation set covers 2008 (366 days), and the test set covers 2009 (365 days). Thus, this time-aware evaluation procedure ensures that all forecasts are strictly based on future data, as would be the case in an actual forecasting situation.

4. Exploratory Data Analysis

4.1 Temperature Time-Series

Figure 1 presents the daily mean temperature time series for all five cities over the entire 10-year period. Strong annual seasonality is present in all cities, with amplitudes of about 20-25°C. Oslo shows the widest temperature range (~-15°C to +25°C), while the three Western European cities, De Bilt, Dusseldorf, and Maastricht, show very similar patterns owing to their geographical proximity.

4.2 Seasonal Decomposition

The additive seasonal decomposition of De Bilt mean temperature in 365 days is shown in Figure 2. The trend component shows a modest warming during the decade. Also, the seasonal component reflects the main annual cycle and the residual component (standard deviation about 3-4 degrees Celsius) indicates the daily variability of weather that the models aim to forecast.

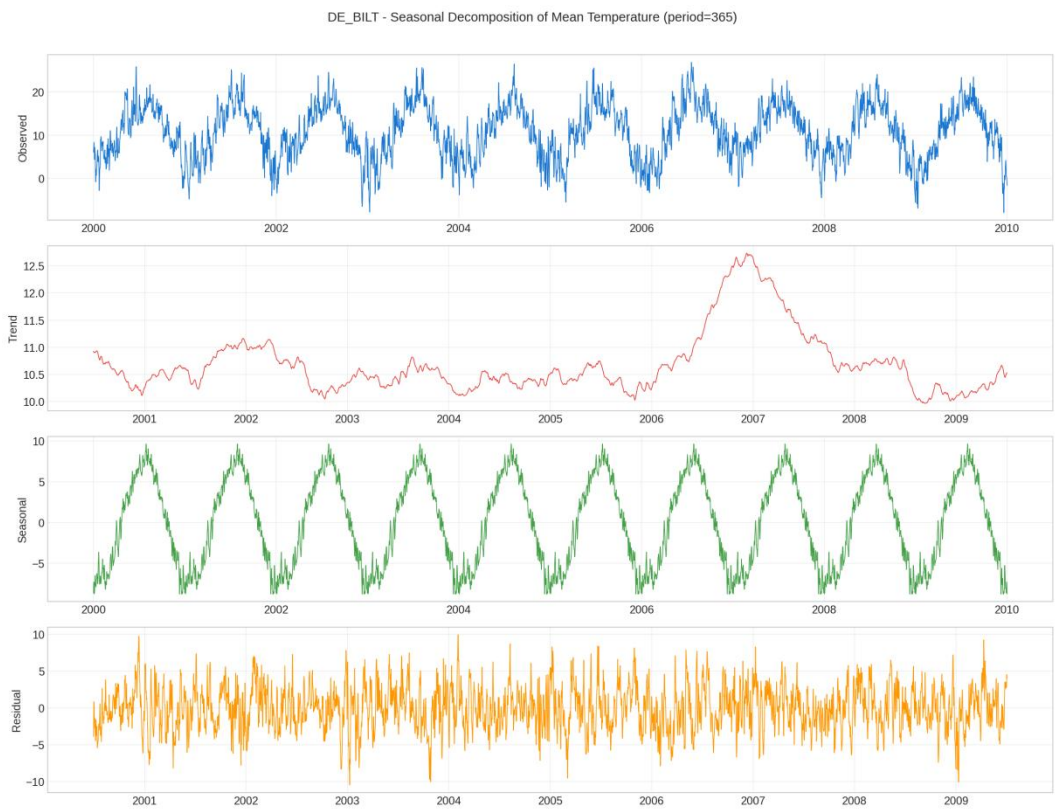

Figure 2: Seasonal decomposition of De Bilt mean temperature (additive, period=365).

4.3 Monthly Temperature Distribution

The boxplots of the monthly temperature have been illustrated in Figure 3 of each city. The temperature varies widely in different seasons. As an example, summer months (June to August) register the lowest interquartile ranges and they are more predictive. But transitional months (March and April, October and November) demonstrate greater variation. It implies that model errors could be heterogeneous by season.

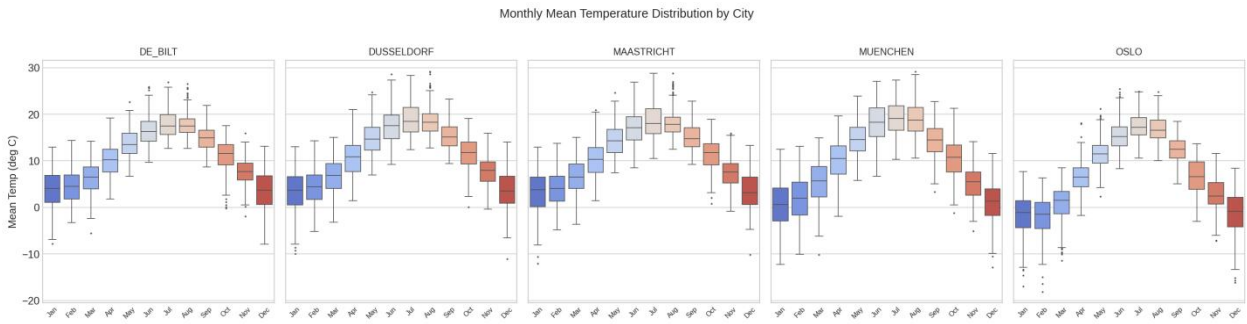

Figure 3: Monthly temperature boxplots by city.

4.4 Feature Correlations

The correlation matrix of the intra-city feature in De Bilt is shown in Figure 4. Temperature features (temp_mean, temp_min, temp_max) are very inter-correlated ($r > 0.95$) and highly correlate with global_radiation ($r=0.7$). The humidity has a

moderate negative correlation with the temperature (r=-0.4). Such correlation patterns are observed in all five cities.

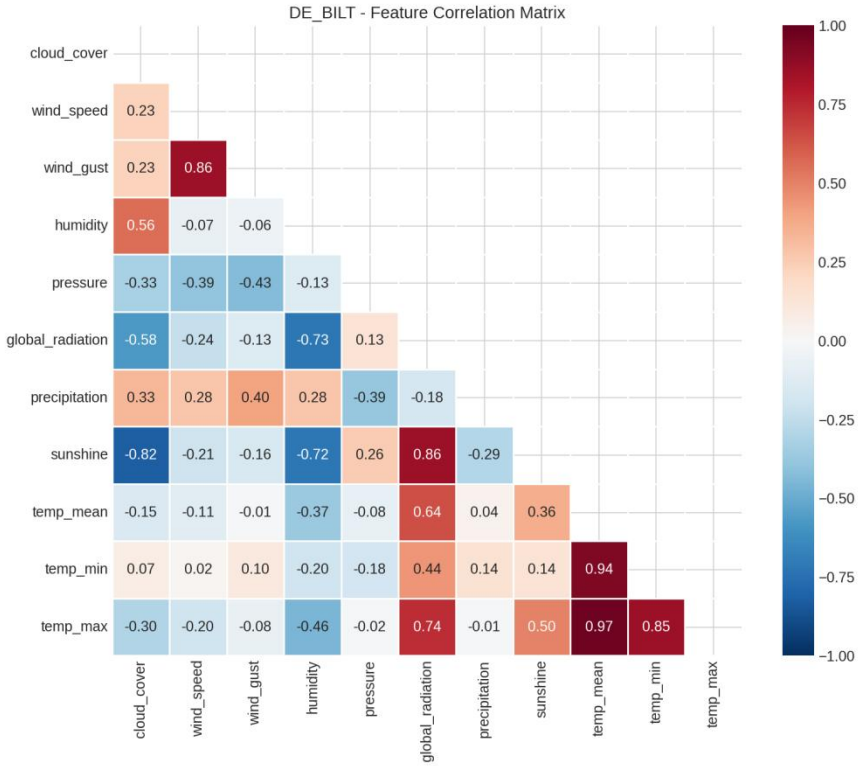

Figure 4: Intra-city feature correlation heatmap (De Bilt).

4.5 Cross-City Temperature Correlation

As shown in Figure 5, three geographically close Western European cities (De Bilt, Dusseldorf and Maastricht) have correlations over 0.97 whereas Oslo and Muenchen have slightly lower yet significant correlations (r = 0.88-0.93) with the cluster. Although there is a high cross-city correlation, we will use a separate model per city so that each model can learn the city-specific weather characteristics.

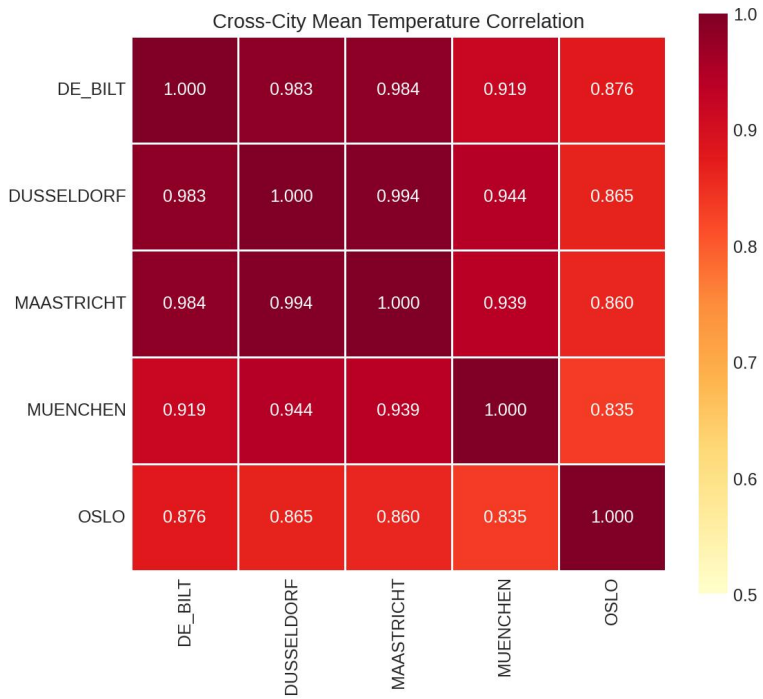

Figure 5: Cross-city mean temperature correlation matrix.

5. Preprocessing and Feature Engineering

5.1 Data Cleaning

There are no null values in the given data set. The date column was transformed into a datetime index. The data set authors substituted less than 5 percent of non-valid entries with the column averages. There were no other imputations.

5.2 Target Variable Construction

In each city, there were three target variables which were generated through a shift in the temperature columns by one day: target_temp_mean, target_temp_min, and target_temp_max. This formulation means that the model will predict the temperature of tomorrow based on the current features (today) without any data leakage.

5.3 Feature Engineering — Four Configurations

In order to assess the role of various feature types in a systematic manner, we have constructed a 2x2 factorial experiment consisting of four feature configurations:

Config	Description	Lag/Roll	Chinese	XGB Features	LSTM Input
A	Raw 11 Only	No	No	11	30-day × 11 features
B	Raw 11 + Chinese	No	Yes	29	30-day × 29 features

C	Raw 11 + Lag/Rolling	Yes	No	132	30-day × 11 features
D	Full (Lag/Roll + Chinese)	Yes	Yes	150	30-day × 29 features

Table 2: Four controlled experiment configurations. The LSTM always receives the same feature set in its sliding window (11 or 29 features); lag/rolling features are only added to XGBoost's tabular input.

5.3.1 Lag and Rolling Features (Configs C and D)

In the case of XGBoost, we develop 7-day lag features of all 11 weather predictors (77 features) and rolling-window statistics (7-day and 30-day means and standard deviations, shifted by 1 day to avoid leakage) of all 11 predictors (44 features). This totals up to 132 tabular features along with the 11 raw features. LSTM is not given explicit lag features since it already has a built-in 30-day sliding window that can capture temporal dependencies.

5.3.2 Chinese Cosmological Features (Configs B and D)

We introduce three groups of features derived from traditional Chinese cosmological systems.
Solar Term (24): The computation of the sun geocentric ecliptic longitude by the use of the PyEphem library. Every date is given a solar term index (0-23) and day in term, as well as cyclical sin/cos encodings. The year is divided into 24 unequal parts by the solar terms which are based on astronomical events (solstices, equinoxes, and intermediate points).

Heavenly Stems and Wuxing (天干/五行): A year is assigned a Heavenly Stem index according to (year − 4) mod 10, which leads to its Wuxing element (Wood, Fire, Earth, Metal, Water) and Yin-Yang polarity. Cyclical sin/cos encodings are added for stem and wuxing indices.

Interaction Features: Three interaction terms encode the traditional Chinese meteorological theory (五运六气) that the same solar term may behave differently depending on the year's element: wuxing × solar_term, polarity × solar_term, and wuxing × polarity × solar_term.

The Chinese features add 18 dimensions: 4 cyclical time encodings (day/month sin/cos), 4 solar term features, and 10 wuxing/stem features.

6. Method

6.1 XGBoost (Gradient Boosting)

XGBoost [4] is an algorithm based on gradient boosting decision trees, suitable for tabular features. For each city and each target variable (average temperature, minimum temperature, maximum temperature), we will train an independent XGBoost regression model. The hyperparameters will be adjusted through grid search on the following parameters: n_estimators ∈ {500, 1000}, max_depth ∈ {5, 7, 9}, learning_rate ∈ {0.01, 0.05, 0.1}, min_child_weight ∈ {3, 5}, while subsample = 0.8 and colsample_bytree = 0.8. The best model for each city-target variable pair will be selected based on the average absolute error of the validation set. Thus, a total of 36 grid search configurations will be generated for each city-target variable pair, and 15 pairs will be evaluated in each experiment.

6.2 Bidirectional LSTM with Attention

The LSTM architecture processes a 30-day sliding window of weather feature data. It consists of a structure with two layers of bidirectional LSTM (with a hidden layer size of 128 and a dropout rate of 0.3), followed by a self-attention mechanism for weighted summation over all time steps, and a 3-layer feedforward head that outputs three concurrent predictions (average temperature, minimum temperature, and maximum temperature). Training uses the AdamW optimizer [5] (with a learning rate of 1e−3 and weight decay of 1e−4), mean squared error loss, ReduceLROnPlateau scheduler (with a factor of 0.5 and patience of 8), a gradient clipping value of 1.0, and a 20-patience early stopping strategy on validation loss. Both features and targets are standardized using the statistics of the training set.

6.3 Feature Importance (XGBoost)

Figure 6 shows the top 20 features sorted by importance in the Deibel temperature prediction (configuration D). The temperature characteristics for the current day dominate: the average temperature, maximum temperature, and minimum temperature for today are the strongest predictors of tomorrow's temperature. The 7-day rolling average temperature feature and the 1-day lag feature also rank highly. The seasonal feature (sine and cosine values of the solar cycle) is among the top 20, but its contribution is smaller compared to the features based on lag time.

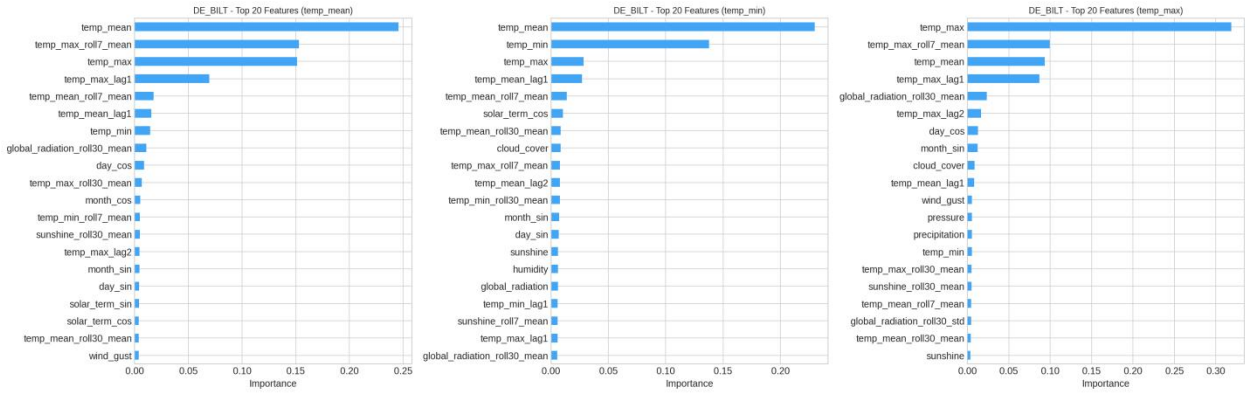

Figure 6: XGBoost top 20 feature importance for De Bilt (Config D: full features).

7. Validation Strategy

We adopted a strict chronological division method for the training/validation/testing sets: the period from 2000 to 2007 served as the training set (approximately 2922 samples), 2008 was the validation set (366 samples), and 2009 was the test set (365 samples). This time-based division method avoids the problem of data leakage that may occur when randomly splitting time series data. The validation set is only used for hyperparameter selection (using XGBoost for grid search to find the best model, using LSTM for early stopping, and learning rate scheduling). All reported metrics were calculated on the 2009 test set, and the model never saw this test set during training or tuning.
By implementing day shifting on rolling window and lag features for at least one day, the potential data leakage issue has been effectively handled without any data pollution from the future data into the current feature set. Meanwhile, during the data preparation process of LSTM, the implementation of LSTM's StandardScaler has been calibrated with the training data.

8. Evaluation and Results

8.1 Metrics

Four metrics are adopted for this project, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and coefficient of determination (R^2). Among these four metrics, MAE and RMSE are the main indicators of model's performance due to their alignment with the actual temperature units and their handling of near-zero values.

Note on MAPE: *As for the MAPE, it has been proven to be unreliable for temperature forecasting since it will generate dynamic fluctuation on percentage error with only a very small amount of absolute error [1]. This disadvantage will frequently appear since the actual temperature could frequently cross 0°C. Given this information, the MAPE failed completely at LSTM models in this project, witnessing error rates over 1,000,000% for 9 out of 15 pairs. Even though the MAPE reported scores at a reasonable range from 37% to 115% for the XGBoost models, its unreasonable error rates shown in the LSTM models has proven this metric as a unreasonable metric. In this case, the MAE and RMSE are used as the primary metrics for this project's model evaluation process.*

8.2 XGBoost Results — All Four Configurations

As it has been shown in the Tables 3 to Tables 6, the XGBoost models were trained with four configurations, including the raw 11 features from the original datasets from Kaggle (Config A), the raw 11 features supplemented with Chinese features (Config B), the raw 11 features supplemented with lag / rolling features (Config C), and the

raw 11 features supplemented with Chinese features and lag / rolling features (Config D). It is notable that the XGBoost models with these four configurations witness stable performance, while their average MAE range from 1.780°C (Config B) to 1.819°C (Config A).

8.2.1 Config A: Raw 11 Features Only

City	Target	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
DE_BILT	temp_mean	1.508	1.914	42.2	0.912
DE_BILT	temp_min	2.002	2.513	75.6	0.815
DE_BILT	temp_max	1.869	2.358	39.0	0.901
DUSSELDORF	temp_mean	1.610	2.110	42.6	0.909
DUSSELDORF	temp_min	2.119	2.822	106.1	0.813
DUSSELDORF	temp_max	2.072	2.608	41.2	0.897
MAASTRICHT	temp_mean	1.648	2.185	37.0	0.901
MAASTRICHT	temp_min	1.796	2.435	67.5	0.849
MAASTRICHT	temp_max	2.118	2.667	43.6	0.889
MUENCHEN	temp_mean	1.625	2.094	68.4	0.934
MUENCHEN	temp_min	1.526	1.986	76.8	0.926
MUENCHEN	temp_max	2.459	3.159	47.3	0.889
OSLO	temp_mean	1.466	1.914	54.3	0.948
OSLO	temp_min	1.439	1.863	82.8	0.941
OSLO	temp_max	2.024	2.515	51.9	0.933
AVERAGE		1.819	2.343	58.4	0.897

Table 3: XGBoost test set results — Config A (11 raw features only).

8.2.2 Config B: Raw 11 + Chinese Features

City	Target	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
DE_BILT	temp_mean	1.440	1.861	39.7	0.916
DE_BILT	temp_min	1.971	2.474	74.9	0.821
DE_BILT	temp_max	1.785	2.269	34.8	0.908
DUSSELDORF	temp_mean	1.548	2.042	35.7	0.915
DUSSELDORF	temp_min	2.161	2.872	108.3	0.806
DUSSELDORF	temp_max	1.996	2.536	39.1	0.902
MAASTRICHT	temp_mean	1.623	2.126	35.4	0.906

MAASTRICHT	temp_min	1.764	2.354	65.9	0.859
MAASTRICHT	temp_max	2.061	2.597	43.5	0.895
MUENCHEN	temp_mean	1.599	2.058	64.8	0.936
MUENCHEN	temp_min	1.481	1.952	78.6	0.928
MUENCHEN	temp_max	2.463	3.112	45.1	0.892
OSLO	temp_mean	1.398	1.830	52.1	0.952
OSLO	temp_min	1.426	1.806	89.7	0.945
OSLO	temp_max	1.987	2.450	51.8	0.936
AVERAGE		1.780	2.289	57.3	0.901

Table 4: XGBoost test set results — Config B (11 raw + Chinese features). Best XGBoost average MAE (1.780°C).

8.2.3 Config C: Raw 11 + Lag/Rolling

City	Target	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
DE_BILT	temp_mean	1.518	1.936	45.1	0.910
DE_BILT	temp_min	1.944	2.445	75.1	0.825
DE_BILT	temp_max	1.834	2.336	38.7	0.903
DUSSELDORF	temp_mean	1.607	2.131	44.4	0.907
DUSSELDORF	temp_min	2.138	2.898	105.5	0.803
DUSSELDORF	temp_max	1.978	2.542	38.6	0.902
MAASTRICHT	temp_mean	1.663	2.202	38.9	0.899
MAASTRICHT	temp_min	1.809	2.551	64.7	0.834
MAASTRICHT	temp_max	2.054	2.619	49.2	0.893
MUENCHEN	temp_mean	1.593	2.040	67.1	0.938
MUENCHEN	temp_min	1.524	1.986	77.5	0.926
MUENCHEN	temp_max	2.406	3.062	42.1	0.895
OSLO	temp_mean	1.420	1.869	42.5	0.950
OSLO	temp_min	1.438	1.861	91.4	0.941
OSLO	temp_max	1.931	2.423	50.0	0.937
AVERAGE		1.790	2.327	58.1	0.897

Table 5: XGBoost test set results — Config C (11 raw + lag/rolling features, no Chinese).

8.2.4 Config D: Full Feature Set (Lag/Roll + Chinese)

City	Target	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
DE_BILT	temp_mean	1.465	1.889	43.6	0.914
DE_BILT	temp_min	1.936	2.443	74.9	0.825
DE_BILT	temp_max	1.826	2.334	39.1	0.903
DUSSELDORF	temp_mean	1.598	2.145	42.2	0.906
DUSSELDORF	temp_min	2.149	2.910	114.5	0.801
DUSSELDORF	temp_max	2.067	2.677	47.1	0.891
MAASTRICHT	temp_mean	1.654	2.197	38.2	0.899
MAASTRICHT	temp_min	1.799	2.519	64.9	0.838
MAASTRICHT	temp_max	2.065	2.627	49.8	0.893
MUENCHEN	temp_mean	1.593	2.024	67.4	0.939
MUENCHEN	temp_min	1.510	1.983	76.5	0.926
MUENCHEN	temp_max	2.419	3.076	44.3	0.894
OSLO	temp_mean	1.388	1.839	43.4	0.952
OSLO	temp_min	1.434	1.859	90.7	0.941
OSLO	temp_max	1.940	2.414	50.3	0.938
AVERAGE		1.790	2.329	59.1	0.897

Table 6: XGBoost test set results — Config D (full feature set: lag/rolling + Chinese).

8.3 LSTM Results — All Four Configurations

The Table 7 to Table 10 demonstrate the LSTM models' performance under the identical set of configurations from the XGBoost models, including the raw 11 features from the original datasets from Kaggle (Config A), the raw 11 features supplemented with chinese features (Config B), the raw 11 features supplemented with lag / rolling features (Config C), and the raw 11 features supplemented with Chinese features and lag / rolling features (Config D). It is remarkable that the best performance comes from the LSTM model with the simplest configuration (Config A) with the average MAE of 2.512°C, while the worst one comes from the most complex configuration (Config D) with the worst average MAE of 2.708°C.

MAPE note: *Due to the unreasonably exceeding MAPE values shown in the 9 out of 15 city-target pairs (see Section 8.1), only 6 city-target pairs, whose MAPE values are less than 1,000%, are shown in the Tables, while the others are marked with N/A†.*

8.3.1 Config A: Raw 11 Features Only

City	Target	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
DE_BILT	temp_mean	1.978	2.602	N/A†	0.837

DE_BILT	temp_min	2.571	3.196	N/A†	0.701
DE_BILT	temp_max	2.270	3.022	46.4	0.837
DUSSELDORF	temp_mean	2.328	3.046	N/A†	0.810
DUSSELDORF	temp_min	2.726	3.683	102.2	0.681
DUSSELDORF	temp_max	2.812	3.557	49.9	0.808
MAASTRICHT	temp_mean	2.364	3.055	N/A†	0.806
MAASTRICHT	temp_min	2.494	3.253	N/A†	0.730
MAASTRICHT	temp_max	2.849	3.593	81.4	0.799
MUENCHEN	temp_mean	2.563	3.168	N/A†	0.849
MUENCHEN	temp_min	2.242	2.907	N/A†	0.841
MUENCHEN	temp_max	3.460	4.293	70.5	0.795
OSLO	temp_mean	2.103	2.752	N/A†	0.892
OSLO	temp_min	2.393	3.091	169.8	0.838
OSLO	temp_max	2.524	3.169	N/A†	0.893
AVERAGE		2.512	3.226	86.7*	0.808

*Table 7: LSTM test set results — Config A (11 raw features). Best LSTM average MAE (2.512°C). *MAPE average of 6 valid pairs only.*

8.3.2 Config B: Raw 11 + Chinese Features

City	Target	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
DE_BILT	temp_mean	1.968	2.524	N/A†	0.846
DE_BILT	temp_min	2.525	3.165	N/A†	0.707
DE_BILT	temp_max	2.376	2.997	54.4	0.840
DUSSELDORF	temp_mean	2.317	3.026	N/A†	0.812
DUSSELDORF	temp_min	2.778	3.640	125.8	0.689
DUSSELDORF	temp_max	2.796	3.557	48.9	0.808
MAASTRICHT	temp_mean	2.630	3.325	N/A†	0.770
MAASTRICHT	temp_min	2.628	3.502	N/A†	0.687
MAASTRICHT	temp_max	3.151	3.861	66.6	0.768
MUENCHEN	temp_mean	2.464	3.153	N/A†	0.851
MUENCHEN	temp_min	2.135	2.848	N/A†	0.847
MUENCHEN	temp_max	3.391	4.256	65.8	0.798

OSLO	temp_mean	2.284	3.011	N/A†	0.871
OSLO	temp_min	2.511	3.217	202.5	0.824
OSLO	temp_max	2.720	3.399	N/A†	0.877
AVERAGE		2.578	3.299	94.0*	0.800

*Table 8: LSTM test set results — Config B (11 raw + Chinese features). *MAPE average of 6 valid pairs only.*

8.3.3 Config C: Raw 11 + Lag/Rolling (LSTM uses 11 raw in window)

City	Target	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
DE_BILT	temp_mean	1.971	2.592	N/A†	0.838
DE_BILT	temp_min	2.560	3.144	N/A†	0.711
DE_BILT	temp_max	2.433	3.134	57.7	0.825
DUSSELDORF	temp_mean	2.386	3.147	N/A†	0.797
DUSSELDORF	temp_min	2.753	3.757	101.5	0.668
DUSSELDORF	temp_max	2.829	3.650	58.8	0.797
MAASTRICHT	temp_mean	2.305	2.985	N/A†	0.814
MAASTRICHT	temp_min	2.469	3.264	N/A†	0.728
MAASTRICHT	temp_max	2.793	3.548	70.3	0.804
MUENCHEN	temp_mean	2.609	3.269	N/A†	0.840
MUENCHEN	temp_min	2.299	2.979	N/A†	0.833
MUENCHEN	temp_max	3.481	4.364	63.5	0.788
OSLO	temp_mean	2.090	2.767	N/A†	0.891
OSLO	temp_min	2.408	3.124	174.2	0.834
OSLO	temp_max	2.396	3.080	N/A†	0.899
AVERAGE		2.519	3.254	87.7*	0.804

*Table 9: LSTM test set results — Config C (LSTM receives 11 raw features in its sliding windows; lag/rolling only affects XGBoost). *MAPE average of 6 valid pairs only.*

8.3.4 Config D: Full Feature Set (Lag/Roll + Chinese)

City	Target	MAE (°C)	RMSE (°C)	MAPE (%)	R^2
DE_BILT	temp_mean	2.059	2.649	N/A†	0.831
DE_BILT	temp_min	2.559	3.177	N/A†	0.705
DE_BILT	temp_max	2.347	3.061	48.8	0.833

DUSSELDORF	temp_mean	2.523	3.357	N/A†	0.769
DUSSELDORF	temp_min	2.897	3.921	119.1	0.639
DUSSELDORF	temp_max	3.073	3.928	77.1	0.765
MAASTRICHT	temp_mean	2.670	3.544	N/A†	0.738
MAASTRICHT	temp_min	2.662	3.640	N/A†	0.662
MAASTRICHT	temp_max	3.304	4.171	113.3	0.729
MUENCHEN	temp_mean	2.609	3.241	N/A†	0.842
MUENCHEN	temp_min	2.251	2.916	N/A†	0.840
MUENCHEN	temp_max	3.564	4.376	82.3	0.786
OSLO	temp_mean	2.568	3.423	N/A†	0.833
OSLO	temp_min	2.687	3.654	196.0	0.773
OSLO	temp_max	2.851	3.647	N/A†	0.858
AVERAGE		2.708	3.514	106.1*	0.774

*Table 10: LSTM test set results — Config D (29 features: 11 raw + Chinese). Worst LSTM performance. *MAPE average of 6 valid pairs only. †MAPE exceeds 10⁶% due to near-zero division; see Section 8.1.*

8.4 Model Comparison Visualisations (Config D)

In general, XGBoost achieves consistently lower MAE across all 15 city-target pairs in every configuration, winning 15 out of 15 comparisons. As it is shown below, the model visualization is between the XGBoost model and the LSTM model under the Config D (the raw 11 features supplemented with Chinese features and lag / rolling features).

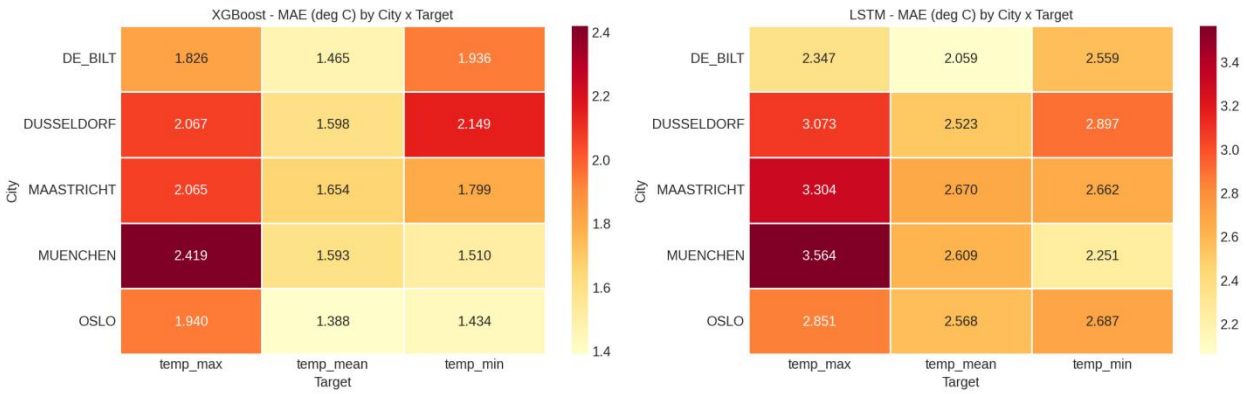

Figure 7: MAE heatmaps — XGBoost (left) vs LSTM (right), Config D.

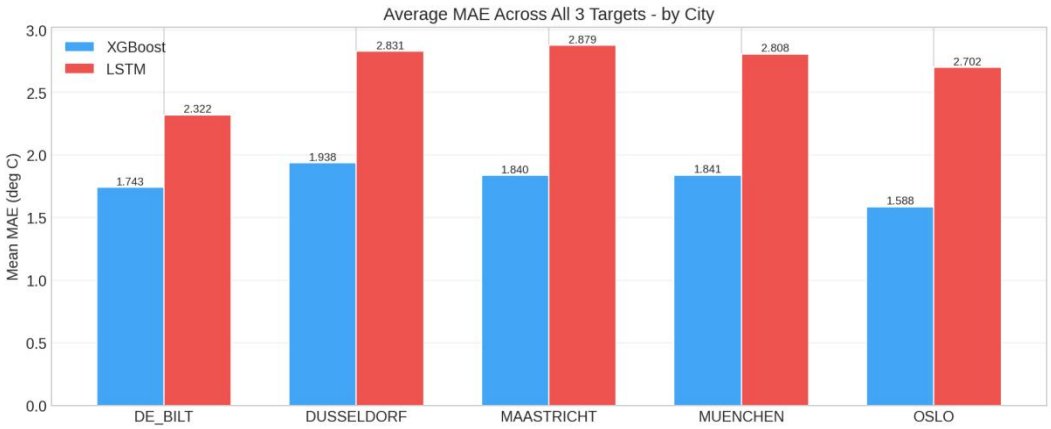

Figure 8: Average MAE by city — XGBoost vs LSTM, Config D.

8.4.1 Prediction vs Actual (De Bilt)

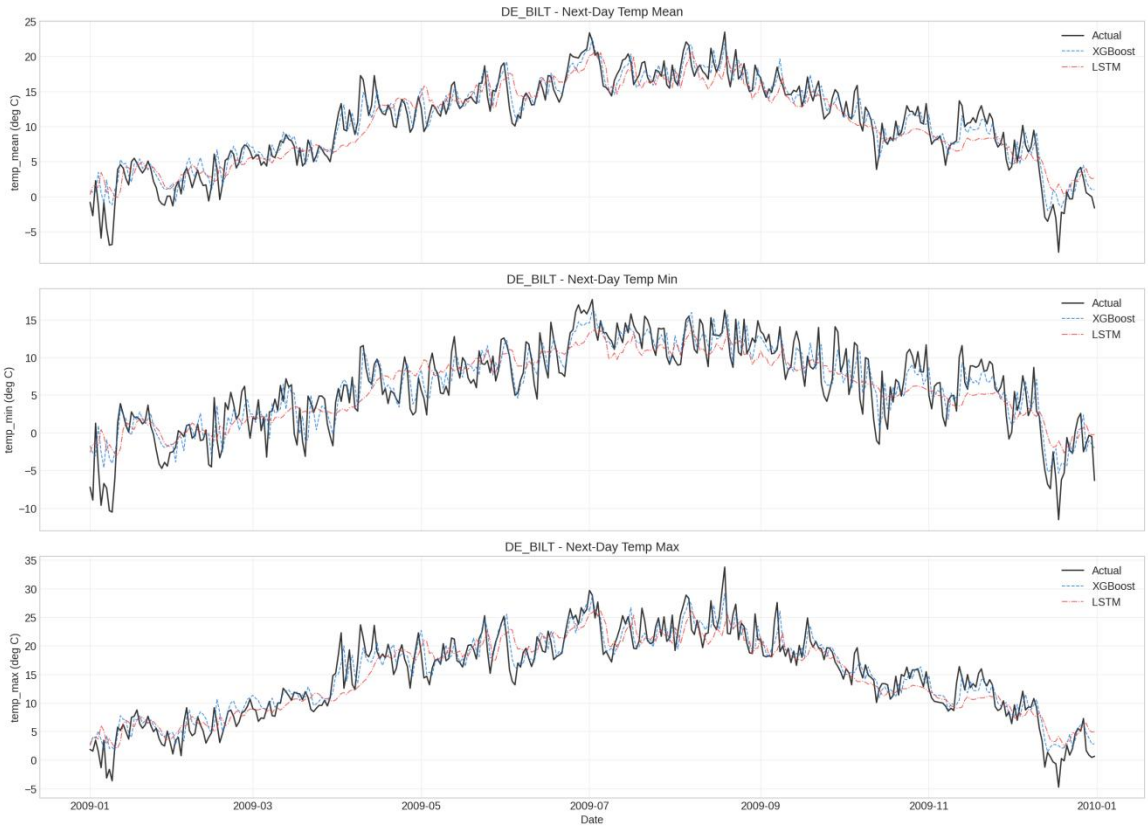

Figure 9: Predicted vs actual temperatures for De Bilt (2009), Config D.

8.4.2 Residual Analysis

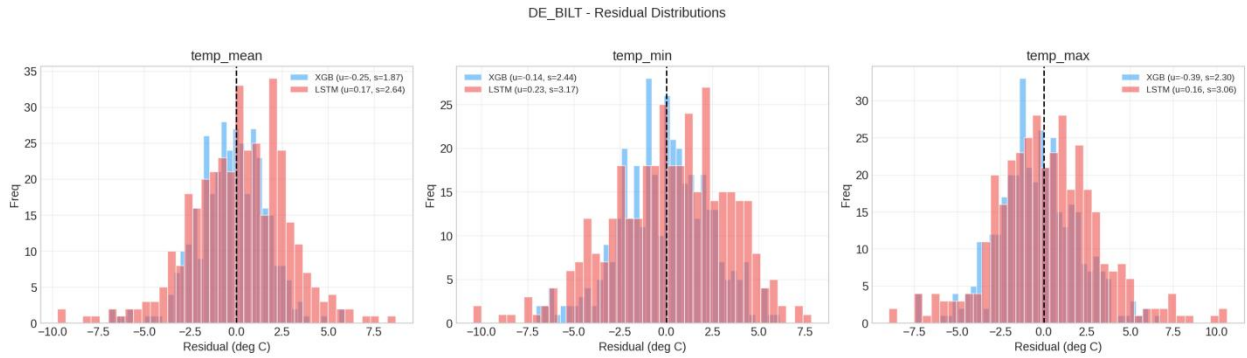

Figure 10: Residual distributions for De Bilt (2009), Config D. XGBoost residuals are more tightly centered around zero.

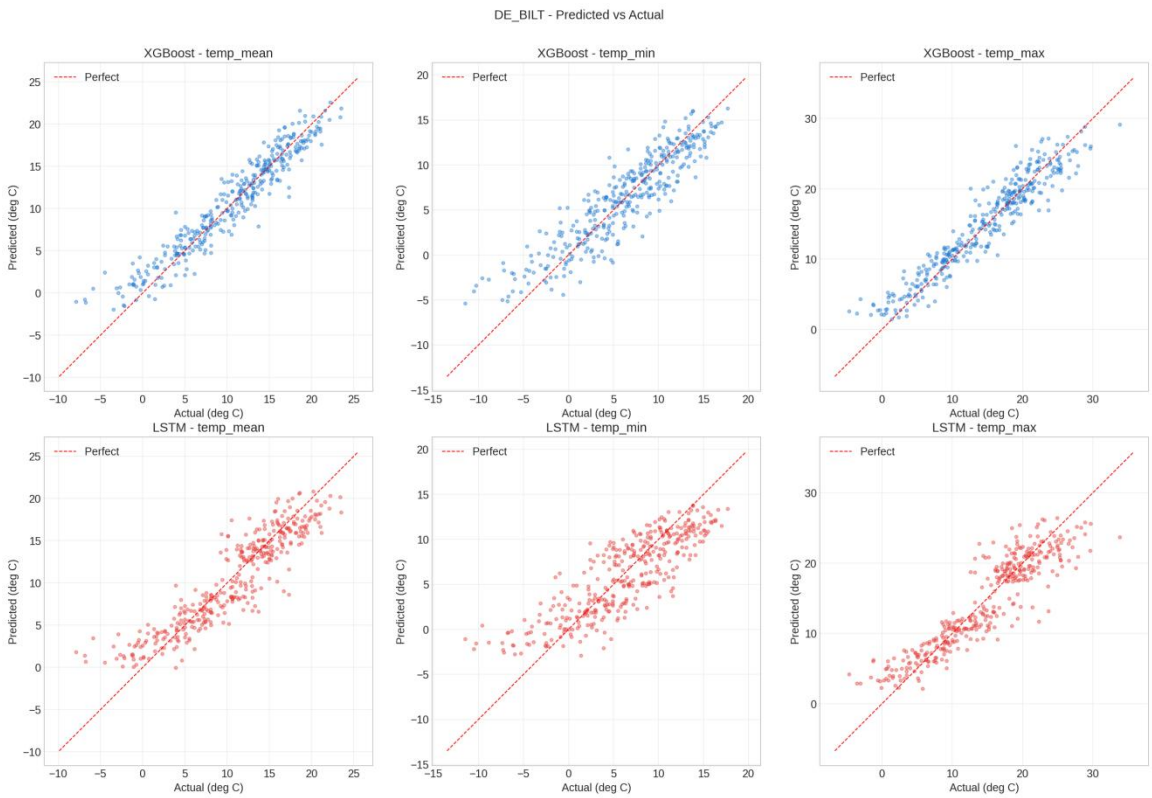

Figure 11: Scatter plots of predicted vs actual values for De Bilt (2009), Config D.

8.5 Controlled Variable Experiment Results

In this project, a 2×2 factorial design is implemented to set controlled variable experiments, where the marginal effect of each feature's engineering strategy can be examined under a controlled environment. The performances of XGBoost and LSTM are shown and compared in the dimensions of MAE and R^2 across all 15 city-target pairs for each configuration.

Configuration	XGB MAE (°C)	LSTM MAE (°C)	XGB R^2	LSTM R^2

A: Raw 11 Only	1.819	2.512	0.897	0.808
B: Raw 11 + Chinese	1.780	2.578	0.901	0.800
C: Lag/Roll (no Chinese)	1.790	2.519	0.898	0.805
D: Lag/Roll + Chinese	1.790	2.708	0.897	0.774

Table 11: Average test set MAE and R^2 across all 15 city-target pairs for each configuration.

8.5.1 Effect of Chinese Features

By comparing the data from Config A and Config B, it is notable that XGBoost improved by −0.039°C (−2.1%), while LSTM worsened by +0.067°C (+2.6%). While XGBoost showed essentially no change (−0.001°C, −0.05%), while LSTM worsened by +0.189°C (+7.5%), under the comparison between Config C and Config D. This difference in the performances of XGBoost and LSTM indicates that Chinese features provide modest XGBoost benefit only when no other temporal features exist, and consistently harm LSTM performance.

8.5.2 Effect of Lag/Rolling Features

- Without Chinese features (A→C): XGBoost MAE decreased by 0.028°C (−1.6%), while LSTM showed no change (+0.007°C).
- With Chinese features present (B→D): XGBoost showed no improvement (+0.009°C), while LSTM worsened by +0.130°C (+5.0%).

➢ The lag/rolling features provide only marginal gains for XGBoost and are neutral to harmful for LSTM.

8.5.3 Statistical Significance (Paired t-Tests)

Since the same 15 city-target pairs appear in all four configurations, we use paired t-tests to remove between-pair variance and detect treatment effects. With 15 matched pairs and $\alpha = 0.05$, the design provides >80% power to detect medium-to-large effects (Cohen's $d \geq 0.8$) [3].

Comparison	Δ MAE (°C)	t-stat	p-value	Sig.	Cohen d
XGB: A→B (+ Chinese)	−0.039	−4.47	0.0005	***	−1.20
XGB: C→D (+ Chinese)	−0.001	−0.11	0.918	ns	−0.03
XGB: A→C (+ Lag/Roll)	−0.028	−2.83	0.013	*	−0.76
XGB: B→D (+ Lag/Roll)	+0.009	+1.03	0.320	ns	+0.28
LSTM: A→B (+ Chinese)	+0.067	+1.96	0.070	ns	+0.53
LSTM: C→D (+ Chinese)	+0.189	+3.77	0.002	**	+1.01
LSTM: A→C (+ Lag/Roll)	+0.007	+0.42	0.680	ns	+0.11

LSTM: B→D (+ Lag/Roll)	+0.130	+5.70	0.0001	***	+1.52

*Table 12: Paired t-test results for controlled variable comparisons (n=15, α=0.05). Negative Δ MAE = improvement. Significance: *** p<0.001, ** p<0.01, * p<0.05, ns = not significant. Note: A = raw features; B = A + Chinese features; C = A + lag/rolling features; D = B + lag/rolling features.*

8.5.4 Interpretation

The statistical tests reveal three key findings:

1. Chinese features significantly improve XGBoost performance only when lag/rolling features are absent (A→B: p=0.0005, d=−1.20). When lag/rolling features are already present (C→D), the improvement is no longer statistically significant (p=0.918). This confirms that Chinese cosmological features—primarily solar terms—encode seasonal information that largely overlaps with standard time-series feature engineering. Consequently, adding both feature sets yields no additional benefit.

2. Chinese features significantly degrade LSTM performance when combined with lag/rolling features (C→D: p=0.002, d=+1.01). The effect is not statistically significant without lag features (A→B: p=0.070). Each additional input dimension increases the risk of overfitting given only ~2,900 training sequences.

3. Lag/rolling features provide modest XGBoost improvement only in isolation (A→C: p=0.013, d=−0.76), but the improvement is no longer significant when Chinese features are already present (B→D: p=0.320). For LSTM, adding lag features on top of Chinese features causes significant degradation (B→D: p<0.001, d=+1.52), representing the strongest effect in the entire experiment.

The overarching conclusion is that both lag/rolling features and Chinese cosmological features encode overlapping seasonal information. Either one alone provides a small XGBoost benefit over raw features, but combining them yields no additional gain. For LSTM, any additional feature dimensions beyond the 11 raw weather variables actively degrade performance due to the curse of dimensionality on a small dataset. While these findings are consistent and statistically robust, several limitations should be considered when interpreting the results.

9. Limitations and Future Work

9.1 Limitations

Limited temporal coverage: The dataset covers 2000–2009 only (10 years). Longer datasets would allow more training data for the LSTM and might change the XGBoost vs LSTM balance.

Single-step forecasting: We only predict 1 day ahead. Multi-step forecasting (3, 7, 14 days) would be more practically useful and may favour different architectures.

Limited city coverage: Only 5 of 18 cities were used due to incomplete feature coverage. Including all 18 cities with imputation strategies could improve generalisability.

MAPE unreliability: MAPE is a required forecasting metric but is fundamentally unsuitable for temperature data due to zero-crossing. We therefore base our conclusions on MAE, reporting MAPE only for compliance.

No external features: Geographic metadata (latitude, altitude, distance from coast) and large-scale climate indices (NAO, ENSO) were not included.

9.2 Future Work

- Evaluating how model advantages change with forecast horizon through multi-step forecasting (3, 7, 14 days ahead).
- Exploring Transformer architectures such as the Temporal Fusion Transformer [2] to better capture long-range dependencies.
- Ensemble stacking of XGBoost and LSTM predictions could combine their complementary strengths.
- Including all 18 cities with missing-feature imputation, and adding geographical metadata (latitude, altitude) as features.
- Testing on more recent data (post-2010) to assess model robustness to climate change trends.
- Subgroup analysis by season or region to investigate whether the effect of solar terms varies across different seasonal contexts.

10. Reference

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